FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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The Patents Rules, 2003

COMPLETE SPECIFICATION

(see section 10)

Blockchain-Based System for Cultural Artifact Provenance Verification and Digital Marketplace Integration

The following specification particularly describes the invention and the manner in which it is to be performed

FIELD OF INVENTION

The present invention relates to the field of distributed ledger technology and digital transaction systems. More particularly, the invention relates to a blockchain-enabled system for generating and managing cryptographic provenance records of cultural artifacts, integrating artifact identity generation, cultural metadata processing, artificial intelligence-based authenticity verification, and a digital marketplace platform for enabling secure and traceable transactions.

BACKGROUND OF THE INVENTION

The global ecosystem of cultural artifacts, artisan crafts, and heritage goods has witnessed significant expansion, driven by increasing consumer demand for authentic, culturally significant products. However, existing technological infrastructures are inadequate for ensuring verifiable authenticity, traceability, and secure transactions of such artifacts.

Several prior art solutions attempt to address aspects of provenance verification and asset tracking: For instance , Australian Patent AU2019100182A4 discloses a system for maintaining provenance of artworks using RFID-based identification linked to artist identity and metadata. However, this system is restricted to artwork domains and does not support broader categories such as artisan crafts or heritage goods. Furthermore, it lacks integration with any transactional marketplace infrastructure.

Another WIPO Patent Application WO2020055926A3 describes a blockchain-based mechanism for establishing provenance of digital assets using cryptographic hashing. While it ensures traceability for digital assets, it does not address physical artifacts, nor does it incorporate cultural metadata, artisan identity linkage, or marketplace interaction layers.

Despite these developments, existing solutions suffer from several technical limitations. There is an absence of a unified system capable of linking physical cultural artifacts with verifiable digital identities. Existing systems further lack integration between provenance verification mechanisms and digital marketplace platforms, thereby preventing seamless transactional workflows. Additionally, such solutions are unable to incorporate cultural metadata, artisan identity validation, and geographic origin verification into provenance systems. They also exhibit limited capability to prevent counterfeit artifacts due to the absence of artificial intelligence-driven authenticity validation mechanisms. Furthermore, there is no provision for automated, trustless transaction execution with enforced economic distribution among stakeholders.

Hence, there exists a need for a comprehensive technical system that enables secure, verifiable provenance of cultural artifacts, integrates authenticity validation, and facilitates decentralized marketplace transactions.

The present invention addresses the above-mentioned problems by providing a unified blockchain-enabled infrastructure that integrates artifact identity generation, provenance recording, artificial intelligence-based authenticity verification, and a digital marketplace within a single system architecture.

OBJECTIVES OF THE INVENTION

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The primary objective of the present invention is to provide a blockchain-enabled system for cultural artifact provenance verification and digital marketplace integration capable of generating a cryptographically verifiable identity for physical cultural artifacts through structured metadata processing and distributed ledger recording. The system seeks to overcome the limitations of existing provenance systems, which either focus on digital assets or lack integration with marketplace infrastructures and cultural validation mechanisms.

Another objective of the invention is to implement a secure artifact identity generation mechanism that constructs a provenance data object from artifact metadata and applies a cryptographic hash function to produce a unique Cultural Provenance Token (CPT), thereby enabling immutable and tamper-resistant artifact identification.

A further objective of the invention is to incorporate an artificial intelligence-based authenticity verification module configured to perform image feature extraction and similarity-based validation, enabling automated assessment of artifact authenticity beyond traditional manual or document-based verification approaches.

The invention also has the objective of providing an integrated cultural metadata framework that enables validation of artisan identity, geographic origin, and cultural classification, thereby ensuring that provenance records are enriched with domain-specific contextual information.

Another objective is to integrate a blockchain-based provenance ledger configured to record artifact registration, ownership transfer, and verification events in an immutable manner, thereby ensuring traceability across the lifecycle of each artifact.

A further objective of the invention is to provide a smart contract-based transaction mechanism capable of executing automated and trustless transactions, including predefined royalty distribution among stakeholders, without reliance on intermediaries.

Another objective of the invention is to provide a unified digital marketplace interface integrated with provenance verification, authenticity validation, and transaction processing modules, enabling seamless interaction between artifact registration, verification, and exchange within a single system architecture..

SUMMARY OF THE INVENTION

The following summary is provided to facilitate a clear understanding of the new features in the disclosed embodiment and it is not intended to be a full, detailed description. A detailed description of all the aspects of the disclosed invention can be understood by reviewing the full specification, the drawing and the claims and the abstract, as a whole.

In one aspect, the present invention provides a blockchain-enabled system for cultural artifact provenance verification, wherein artifact metadata associated with physical cultural goods is processed to generate a cryptographic identity recorded on a distributed ledger.

In one aspect, the system comprises an artisan registration module configured to perform multi-stage identity verification, geographic consistency validation, and generation of an Artisan Identity Token associated with a verified artisan profile.

In one aspect, the invention includes an artifact identity generation module adapted to construct a Provenance Data Object from artifact metadata including artisan identity, geographic origin, cultural classification, material composition, manufacturing technique, and timestamp, and to apply a cryptographic hash function to generate a unique Cultural Provenance Token.

In one aspect, the system further comprises a blockchain provenance ledger configured to record Cultural Provenance Token registration, ownership transfer events, and authenticity verification events, thereby ensuring immutable and decentralized storage of provenance records.

In one aspect, the invention includes an artificial intelligence-based authenticity verification module configured to perform image feature extraction and similarity-based comparison to determine authenticity of artifacts using stored reference data.

In one aspect, the system comprises a smart contract module configured to execute marketplace transactions, wherein ownership transfer, payment processing, and predefined royalty distribution are performed automatically based on encoded transaction conditions.

In one aspect, the invention includes a digital marketplace interface configured to enable artifact discovery, search, display, and transaction processing, integrated with provenance verification and authenticity validation mechanisms.

In one aspect, the present invention addresses the technical problem of enabling reliable, tamper-resistant verification and traceability of physical cultural artifacts across distributed environments by introducing a unified architecture combining cryptographic identity generation, blockchain-based recording, and artificial intelligence-driven validation.

In one aspect, the invention provides the technical advantage of improved transparency, traceability, and automation over existing systems by integrating provenance verification, authenticity assessment, and transaction execution within a single platform.

In one aspect, the invention enables scalable deployment of a decentralized cultural artifact verification and trading infrastructure, supporting secure and verifiable interactions without dependence on centralized intermediaries or isolated verification mechanisms.

BRIEF DESCRIPTION OF THE DRAWINGS

The manner in which the present invention is formulated is given a more particular description below, briefly summarized above, may be had by reference to the components, some of which is illustrated in the appended drawing It is to be noted; however, that the appended drawing illustrates only typical embodiments of this invention and are therefore should not be considered limiting of its scope, for the system may admit to other equally effective embodiments.

Throughout the drawings, the same drawing reference numerals will be understood to refer to the same elements and features.

The features and advantages of the present invention will become more apparent from the following detailed description a long with the accompanying figures, which forms a part of this application and in which:

FIG. 1 illustrates a system architecture of a cultural verification platform (100)

FIG. 2 illustrates a blockchain provenance architecture depicting a blockchain gateway module

FIG. 3 illustrates an artisan registration workflow executed by the artisan registration module

FIG. 4 illustrates an artifact identity generation workflow executed by the artifact identity generator

FIG. 5 illustrates a blockchain recording workflow

FIG. 6 illustrates an authenticity verification workflow

FIG. 7 illustrates a marketplace transaction workflow

FIG. 8 illustrates an artificial intelligence classification and recommendation architecture

REFERENCE NUMERALS

100	a system architecture of a cultural verification platform

DETAILED DESCRIPTION OF THE INVENTION

The present invention will now be described in detail with reference to the accompanying drawing. The following description is provided to illustrate the invention and should not be construed as limiting the scope of the invention.

The present invention provides a unified platform (100) configured to enable end-to-end management of cultural artifacts through integrated processes of artisan registration, artifact identity generation, provenance recording, authenticity verification, and transaction execution. In operation, the platform (100) initially receives artisan information and performs identity validation and geographic consistency verification before generating and storing a verified artisan record. Subsequently, the platform (100) receives artifact metadata and images, processes the images using an artificial intelligence-based classification functionality, and constructs a structured data object comprising artifact and artisan information. A cryptographic hashing operation is applied to generate a unique provenance identifier, which is recorded on a blockchain ledger while associated data is stored in a database and distributed storage system. The platform (100) further enables authenticity verification by comparing newly submitted artifact data with stored feature representations and recording verification outcomes. Additionally, the platform (100) facilitates marketplace transactions through a smart contract-based mechanism that executes payment processing, ownership transfer, and automated distribution of transaction value upon satisfaction of predefined conditions. Through this integrated workflow, the platform (100) ensures secure, traceable, and verifiable lifecycle management of cultural artifacts

Referring to FIG. 1, a cultural verification platform (100) is illustrated as an integrated system comprising core processing components and artificial intelligence, commerce, and blockchain components.

The platform (100) includes a registration functionality through which artisan-related information is received and processed. The platform (100) further includes an identity generation functionality that receives validated inputs and generates a unique identity corresponding to an artifact. The platform (100) stores artisan-related data and artifact-related data within a structured database maintained as part of the system.

The platform (100) further includes a ledger functionality configured to record provenance-related data in an immutable manner. The platform (100) also includes a marketplace functionality through which artifact data is presented for interaction and transaction.

Additionally, the platform (100) includes an artificial intelligence-based classification functionality configured to process artifact images and generate classification outputs. These outputs are utilized within the platform (100) for both recommendation and verification purposes. The platform (100) further includes a recommendation functionality configured to generate personalized outputs based on classification data.

The platform (100) also includes a transaction execution functionality configured to perform purchase-related operations. The interaction between all components of the platform (100) enables continuous flow of data from registration, identity generation, storage, verification, and transaction stages.

Referring to FIG. 2, the platform (100) includes a blockchain interaction functionality configured to handle transaction generation and recording.

The platform (100) generates signed transaction data corresponding to different operational events. These events include artifact registration, ownership transfer, and verification events. Each transaction is processed and recorded in a distributed ledger maintained as part of the platform (100).

The platform (100) maintains on-chain data comprising a unique provenance identifier, an artisan identity reference, and a content reference corresponding to stored artifact data.

The platform (100) further includes a distributed storage functionality for storing artifact images and extended documentation. The stored data is linked to the on-chain record through a reference identifier, ensuring that large data objects are maintained outside the ledger while still being verifiable through the ledger.

The architecture enables separation of on-chain and off-chain data while maintaining cryptographic linkage between them.

Referring to FIG. 3, the platform (100) performs a structured artisan registration process.

Initially, the platform (100) receives a registration request along with identity-related inputs. The platform (100) validates the received identity information through an external verification process.

Upon successful validation, the platform (100) performs a geographic consistency check by comparing submitted location data with predefined regional information. If the location data is inconsistent, the process is terminated.

If the location data is consistent, the platform (100) evaluates submitted portfolio information. Based on the evaluation, the platform (100) determines whether the artisan satisfies predefined validation criteria.

Upon successful validation, the platform (100) generates a unique identity associated with the artisan and stores the artisan record in the database. The platform (100) then assigns a verified status to the artisan profile, enabling further participation within the system.

The platform (100) receives artifact-related inputs including metadata and images. The platform (100) processes the images using an artificial intelligence-based classification functionality to extract feature information and determine classification characteristics.

The platform (100) constructs a structured data object comprising artisan-related information, cultural classification, geographic information, material details, creation information, and classification outputs.

The platform (100) applies a cryptographic hashing operation to the constructed data object to generate a unique hash value. The hash value is combined with identity-related information and system-generated information to produce a unique provenance identifier corresponding to the artifact.

The platform (100) stores artifact-related data in a database and distributed storage, while simultaneously preparing the generated identifier for recording in the ledger.

Referring to FIG. 5, the platform (100) performs recording of provenance data on the ledger. The platform (100) receives data corresponding to artifact identity and associated parameters. The platform (100) generates a transaction and applies a digital signing process to ensure authenticity. The signed transaction is broadcast to a distributed validation network. The platform (100) monitors whether consensus is achieved. If consensus is not achieved, the transaction is retransmitted. Upon successful consensus, the platform (100) records the transaction in the ledger. A confirmation output including transaction-related details is generated.

The platform (100) updates the database with a reference linking stored data to the ledger entry. The platform (100) then publishes the artifact information to the marketplace functionality.

Referring to FIG. 6, the platform (100) performs authenticity verification. The platform (100) receives a verification request associated with a provenance identifier. The platform (100) retrieves corresponding data from the ledger and associated data from the database. The platform (100) receives artifact images for verification. The platform (100) processes the images using the artificial intelligence-based classification functionality to extract feature information.

The platform (100) performs a similarity comparison between extracted feature information and previously stored feature information. Based on the similarity value, the platform (100) determines whether the artifact is authentic. If the similarity value meets a predefined condition, the platform (100) confirms authenticity. Otherwise, the platform (100) initiates a manual evaluation process. The platform (100) records the verification outcome in the ledger and provides an authenticity result.

Referring to FIG. 7, the platform (100) performs transaction execution. The platform (100) enables a user to access artifact information and review associated provenance data. The platform (100) enables connection of a digital wallet for transaction processing. The platform (100) generates a transaction contract comprising artifact identity, participant details, transaction value, and predefined distribution parameters.

The platform (100) receives payment into a controlled transaction environment. The platform (100) monitors fulfillment conditions including shipment confirmation, delivery confirmation, and user acceptance. Upon satisfaction of conditions, the platform (100) executes an automated transaction. The platform (100) distributes the transaction value according to predefined proportions and records ownership transfer in the ledger.

The platform (100) generates a digital record confirming the transaction.Referring to FIG. 8, the platform (100) includes an artificial intelligence-based processing architecture.

The platform (100) receives image inputs and textual inputs. The image inputs are processed to extract feature representations. The textual inputs are processed to generate embedded representations. The platform (100) combines image-based and text-based representations using a fusion process to generate a unified feature representation. The platform (100) uses the unified representation to perform classification, scoring, and recommendation operations. The platform (100) generates outputs including classification labels, confidence values, authenticity scores, and personalized recommendations.

In an alternative embodiment, the platform (100) is configured such that the provenance identifier generated for each artifact is implemented as a transferable digital token within the blockchain environment, wherein ownership of the digital token corresponds to ownership of the associated physical artifact. In this embodiment, the platform (100) further enables physical-to-digital linkage through the use of machine-readable authentication elements associated with the artifact, allowing retrieval of provenance data directly from the platform (100). Additionally, the platform (100) may be configured to integrate with external certification systems and registry databases for validating artisan identity and cultural origin information, thereby enhancing the reliability of the stored provenance data. Further, the platform (100) may operate with decentralized identity management mechanisms wherein artisan credentials are verified without centralized storage, while maintaining interoperability with the existing provenance recording and marketplace functionalities.

The best mode of working of the present invention is described with reference to the integrated operation of the platform (100) across registration, identity generation, verification, and transaction stages.

Initially, the platform (100) receives artisan registration data and performs identity validation through a verification interface. Upon successful validation, the platform (100) performs geographic consistency checking and evaluates submitted portfolio information. Once validated, the platform (100) generates a unique identity associated with the artisan and stores the corresponding record.

Subsequently, the platform (100) receives artifact-related data including images and metadata. The platform (100) processes the images using an artificial intelligence-based classification functionality to extract feature representations and generate classification outputs. A structured data object is constructed using artisan identity information, cultural classification, geographic information, material details, and classification outputs.

The platform (100) applies a cryptographic hashing operation to the structured data object to generate a unique provenance identifier. The generated identifier is combined with identity-related information to form a provenance token corresponding to the artifact.

The platform (100) then generates a transaction and records the provenance token in a blockchain ledger through a consensus-based validation process. Simultaneously, artifact images and extended metadata are stored in a distributed storage system, and corresponding references are linked to the blockchain record.

During operation, the platform (100) enables authenticity verification by receiving a verification request and comparing newly submitted artifact images with stored feature representations using a similarity computation. Based on the result, the platform (100) determines authenticity and records the verification outcome in the ledger.

For transaction processing, the platform (100) enables a user to access artifact data, connect a digital wallet, and initiate a transaction. The platform (100) executes the transaction through a smart contract mechanism, wherein payment is held in a controlled environment until predefined conditions are satisfied. Upon confirmation, the platform (100) performs automated distribution of funds and records ownership transfer in the blockchain ledger. The platform (100) then generates a digital record confirming ownership and updates associated data records.

This integrated operation represents the most effective implementation of the invention, ensuring secure provenance recording, reliable authenticity verification, and automated transaction execution.

The present invention provides a unified technical system that enables secure and verifiable management of cultural artifacts by integrating identity generation, provenance recording, authenticity verification, and transaction execution within a single platform. By combining blockchain-based recording with artificial intelligence-based classification and distributed storage, the invention ensures immutability, traceability, and reliability of artifact-related data across its lifecycle.

Additionally, the invention enhances operational efficiency by enabling automated and trustless transaction execution without reliance on intermediaries. The integration of structured metadata processing, similarity-based verification, and decentralized storage mechanisms reduces the likelihood of counterfeit artifacts and ensures consistent validation of cultural and geographic attributes, thereby improving system robustness and data integrity.

CLAIMS

I/We Claim:

1. A cultural artifact verification system comprising:

a platform (100) including an artisan registration module (100) configured to perform multi-stage identity verification and geographic consistency validation and to generate an artisan identity token;

an artifact identity generator (100) configured to construct a provenance data object from artifact metadata including artisan identity, geographic data, cultural classification, material composition, manufacturing attributes, and timestamp, and to apply a cryptographic hash function to generate a cultural provenance token;

a blockchain provenance ledger (100) configured to record the cultural provenance token, ownership transfer data, and verification event data;

a cultural metadata database (100) configured to store artifact metadata, feature representations, and ledger reference identifiers;

an artificial intelligence classifier (100) configured to extract feature vectors from artifact images and perform similarity computation; and

a smart contract module (100) configured to execute transaction logic and record ownership transfer on the blockchain provenance ledger.

2. The system as claimed in claim 1, wherein the artisan registration module (100) is further configured to validate identity data through an external verification interface and to perform geographic consistency checking using a predefined regional dataset.
3. The system as claimed in claim 1, wherein the artifact identity generator (100) is configured to generate the cultural provenance token by concatenating the hash value with the artisan identity token and a system-generated identifier.

4. The system as claimed in claim 1, wherein the blockchain provenance ledger (100) is configured to store transaction data corresponding to artifact registration, ownership transfer, and authenticity verification.
5. The system as claimed in claim 1, wherein the cultural metadata database (100) is configured to maintain a linkage between stored metadata and blockchain records through a reference identifier.
6. The system as claimed in claim 1, wherein the artificial intelligence classifier (100) is configured to process image inputs using a feature extraction model and to generate feature vectors representing artifact characteristics.
7. The system as claimed in claim 1, wherein the artificial intelligence classifier (100) is further configured to perform similarity computation between stored feature vectors and newly generated feature vectors to determine correspondence.
8. The system as claimed in claim 1, further comprising a distributed storage module (100) configured to store artifact images and associated data, wherein stored data is linked to the blockchain provenance ledger through a content reference identifier.
9. The system as claimed in claim 1, wherein the smart contract module (100) is configured to execute transaction processing through an escrow mechanism and to perform automated distribution based on predefined parameters.
10. A method for cultural artifact verification comprising:

receiving artisan data and performing identity validation and geographic consistency verification using a platform (100);

generating an artisan identity token;

receiving artifact metadata and images;

processing the images to extract feature vectors;

constructing a provenance data object and applying a cryptographic hash function to generate a cultural provenance token;

recording the cultural provenance token on a blockchain provenance ledger (100);

storing metadata and feature representations in a database (100);

performing similarity computation between feature vectors; and

recording transaction and verification data on the blockchain provenance ledger (100)

400

Blockchain-Based System for Cultural Artifact Provenance Verification and Digital Marketplace Integration

ABSTRACT

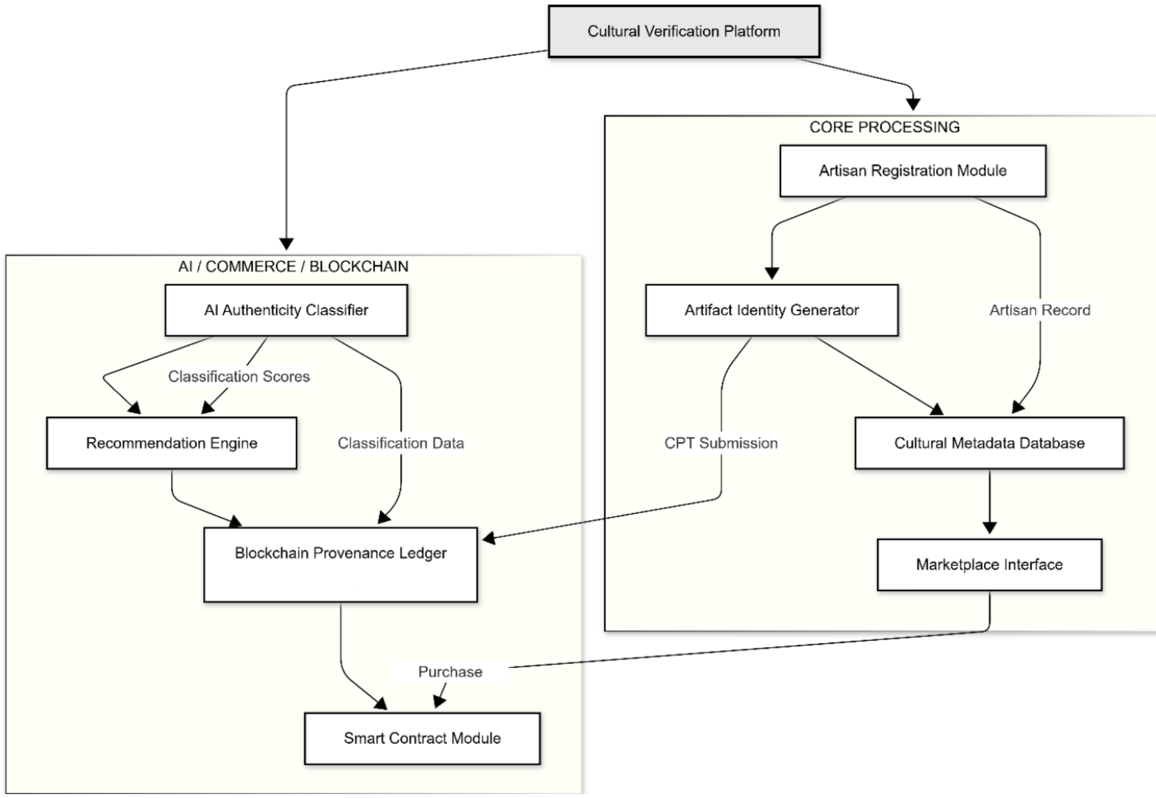
The present invention relates to a cultural artifact verification system comprising a platform (100) including an artisan registration module (100), an artifact identity generator (100), a cultural metadata database (100), a blockchain provenance ledger (100), an artificial intelligence classifier (100), and a smart contract module (100). The artisan registration module (100) performs identity verification and generates an artisan identity token. The artifact identity generator (100) constructs a provenance data object from artifact metadata and applies a cryptographic hash function to generate a cultural provenance token. The blockchain provenance ledger (100) records provenance, ownership transfer, and verification data. The artificial intelligence classifier (100) extracts feature vectors from artifact images and performs similarity computation. The cultural metadata database (100) maintains metadata and reference linkage. The smart contract module (100) executes transaction processing and records ownership transfer on the blockchain provenance ledger (100).

Fig 1

Date: Signature of Applicant

100

DRAWINGS



430

Figure. 1

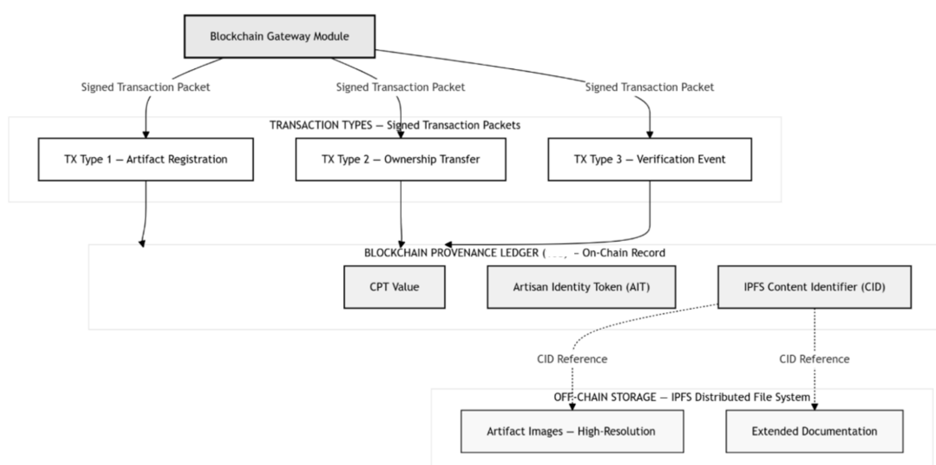


Figure. 2

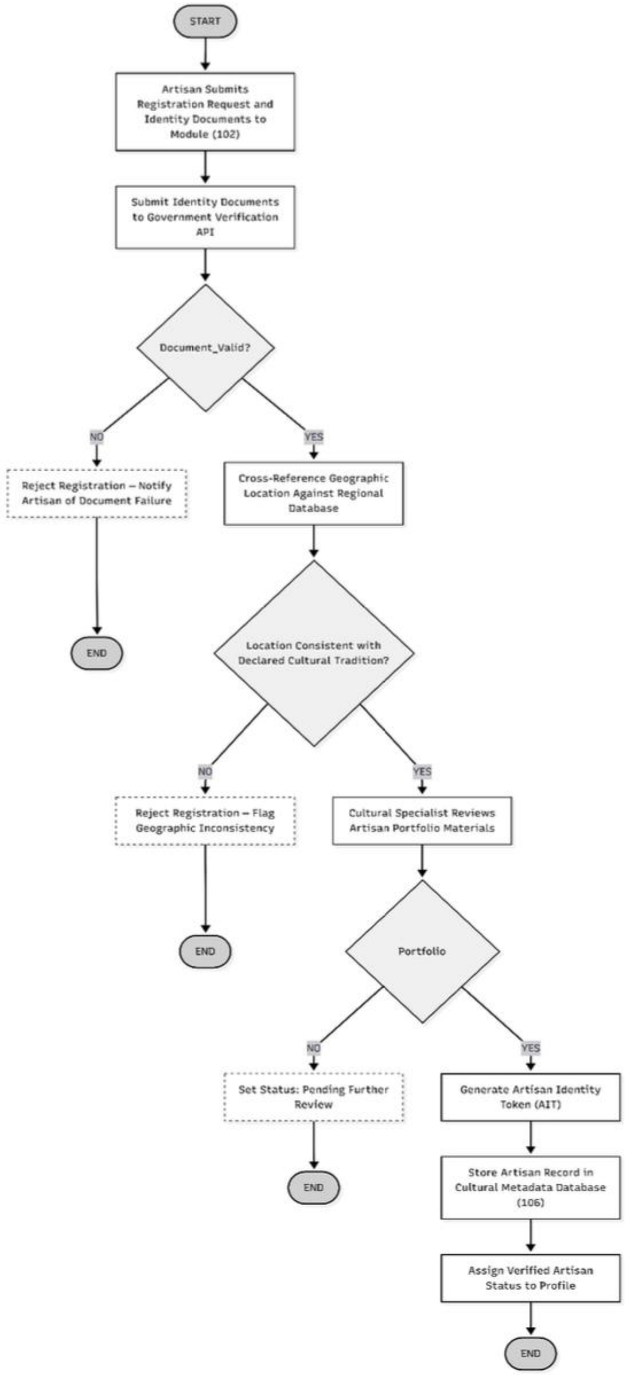

START
Artisan Submits Registration Request and Identity Documents to Module (102)
Submit Identity Documents to Government Verification API
Document_Valid?
NO
YES
Reject Registration – Notify Artisan of Document Failure
Cross-Reference Geographic Location Against Regional Database
END
Location Consistent with Declared Cultural Tradition?
NO
YES
Reject Registration – Flag Geographic Inconsistency
Cultural Specialist Reviews Artisan Portfolio Materials
END
Portfolio
NO
YES
Set Status: Pending Further Review
Generate Artisan Identity Token (AIT)
END
Store Artisan Record in Cultural Metadata Database (106)
Assign Verified Artisan Status to Profile
END

Figure. 3

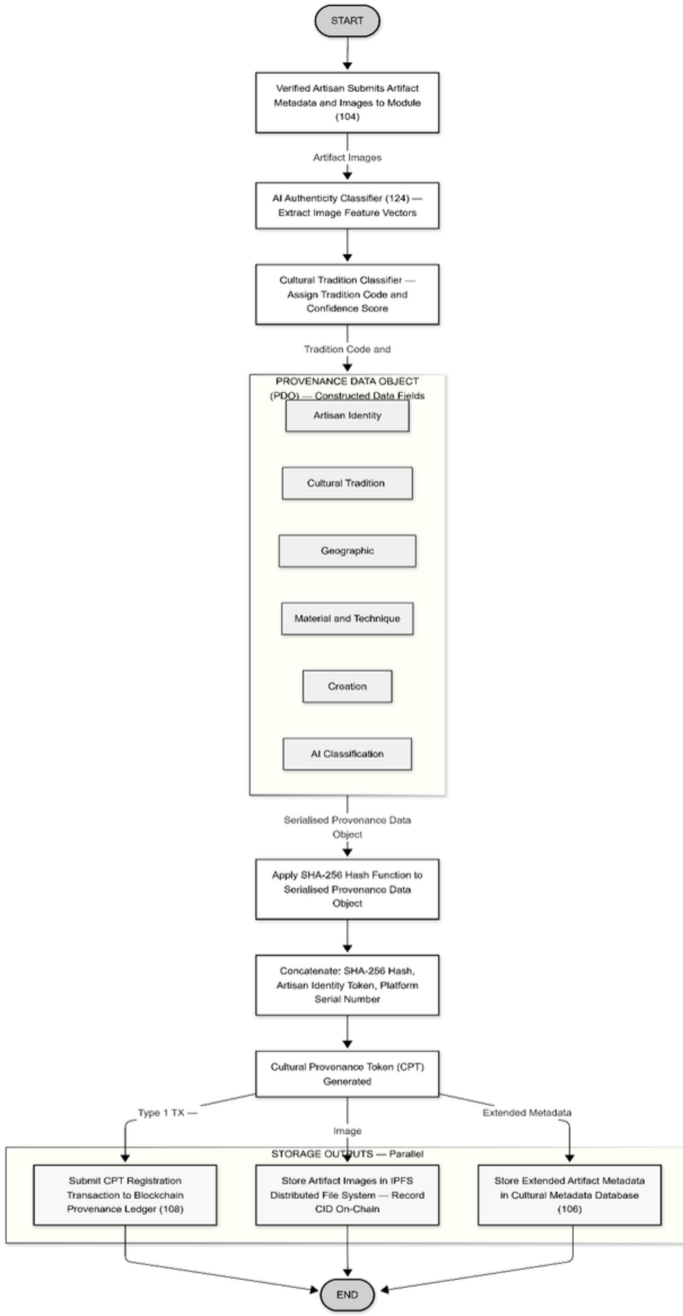

START
Verified Artisan Submits Artifact Metadata and Images to Module (104)
Artifact Images
AI Authenticity Classifier (124) — Extract Image Feature Vectors
Cultural Tradition Classifier — Assign Tradition Code and Confidence Score
Tradition Code and
PROVENANCE DATA OBJECT (PDO) — Constructed Data Fields
Artisan Identity
Cultural Tradition
Geographic
Material and Technique
Creation
AI Classification
Serialised Provenance Data Object
Apply SHA-256 Hash Function to Serialised Provenance Data Object
Concatenate: SHA-256 Hash, Artisan Identity Token, Platform Serial Number
Cultural Provenance Token (CPT) Generated
Type 1 TX —
Image
Extended Metadata
STORAGE OUTPUTS — Parallel
Submit CPT Registration Transaction to Blockchain Provenance Ledger (108)
Store Artifact Images in IPFS Distributed File System — Record CID On-Chain
Store Extended Artifact Metadata in Cultural Metadata Database (106)
END

Figure. 4

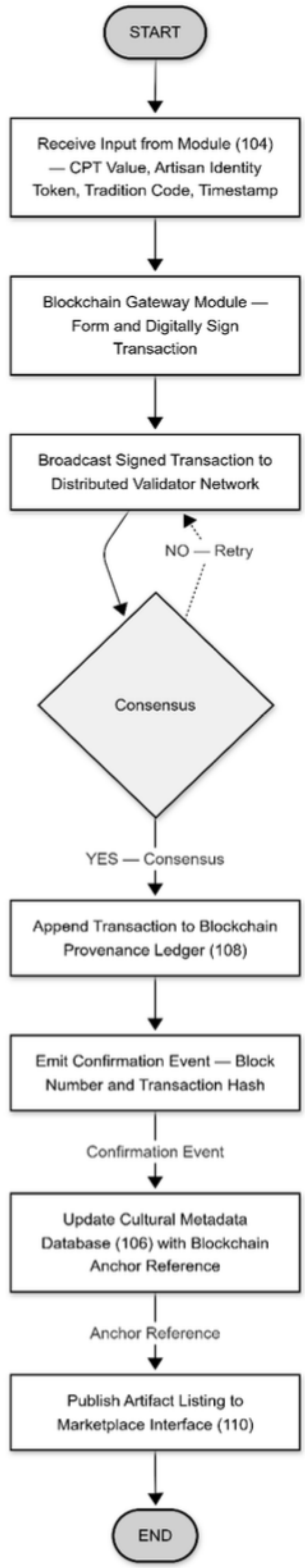
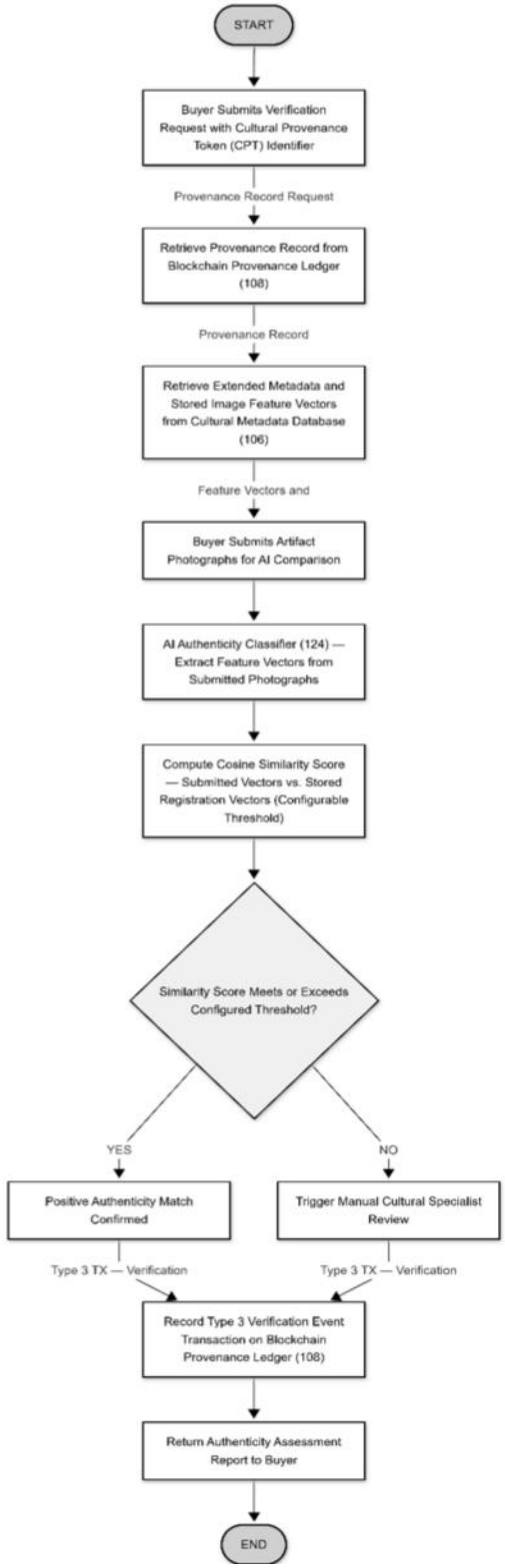

START
Receive Input from Module (104) — CPT Value, Artisan Identity Token, Tradition Code, Timestamp
Blockchain Gateway Module — Form and Digitally Sign Transaction
Broadcast Signed Transaction to Distributed Validator Network
NO — Retry
Consensus
YES — Consensus
Append Transaction to Blockchain Provenance Ledger (108)
Emit Confirmation Event — Block Number and Transaction Hash
Confirmation Event
Update Cultural Metadata Database (106) with Blockchain Anchor Reference
Anchor Reference
Publish Artifact Listing to Marketplace Interface (110)
END

Figure. 5



440 **Figure. 6**

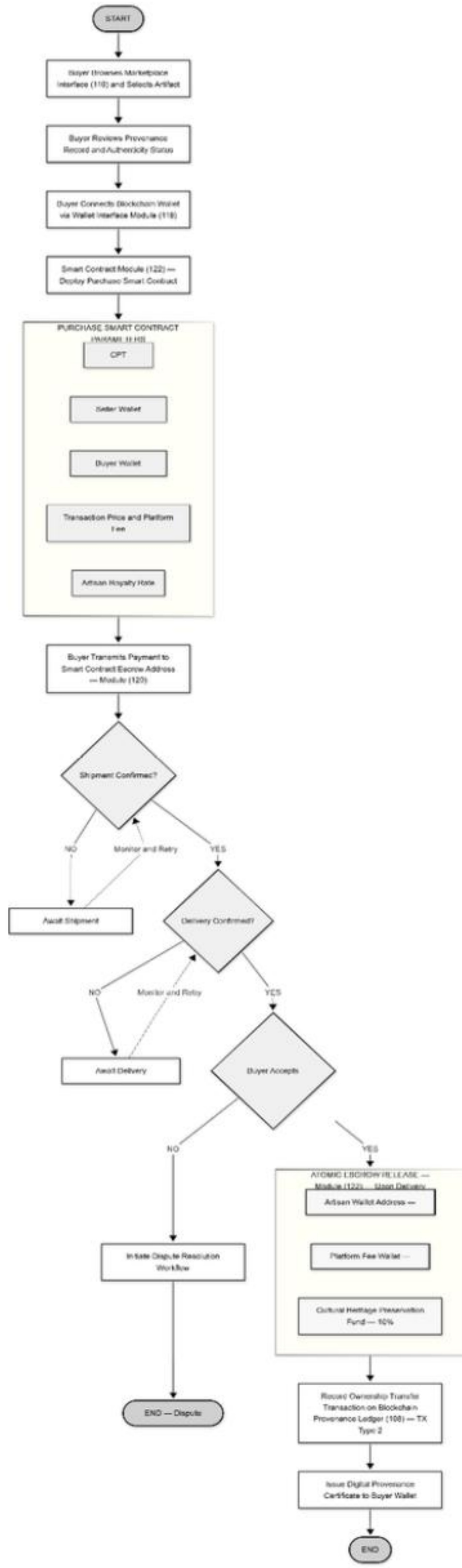
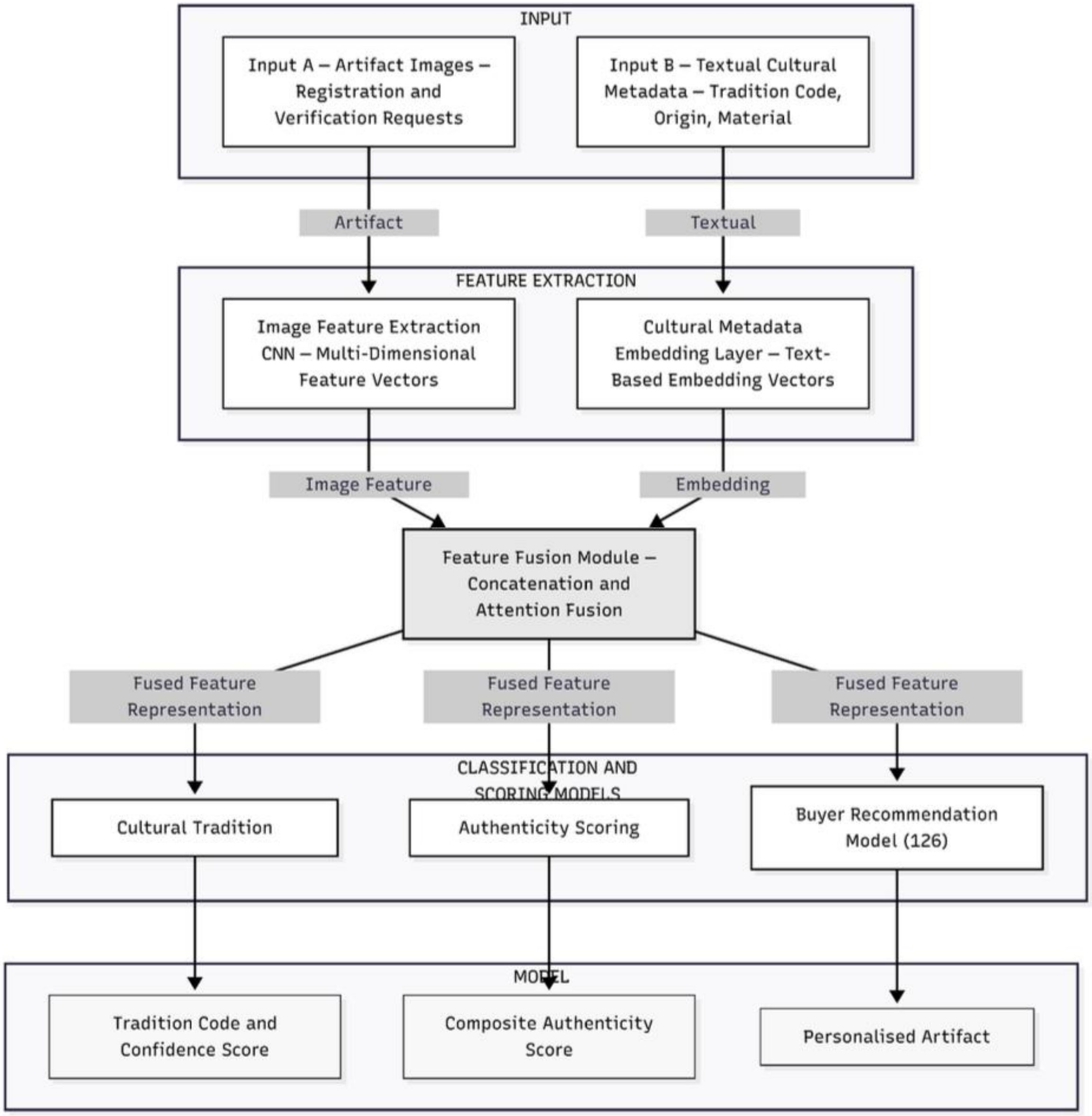

START
Buyer Browses Marketplace Interface (110) and Selects Artifact
Buyer Reviews Provenance Record and Authenticity Status
Buyer Connects Blockchain Wallet via Wallet Interface Module (118)
Smart Contract Module (122) — Deploy Purchase Smart Contract
PURCHASE SMART CONTRACT
CPT
Seller Wallet
Buyer Wallet
Transaction Price and Platform Fee
Artisan Royalty Rate
Buyer Transmits Payment to Smart Contract Escrow Address — Module (120)
Shipment Confirmed?
NO
Monitor and Retry
YES
Await Shipment
Delivery Confirmed?
NO
Monitor and Retry
YES
Await Delivery
Buyer Accepts
NO
YES
Initiate Dispute Resolution Workflow
END — Dispute
ATOMIC ESCROW RELEASE —
Artisan Wallet Address —
Platform Fee Wallet —
Cultural Heritage Preservation Fund — 10%
Record Ownership Transfer Transaction on Blockchain Provenance Ledger (108) — TX Type 2
Issue Digital Provenance Certificate to Buyer Wallet
END

Figure. 7


INPUT
Input A – Artifact Images – Registration and Verification Requests
Input B – Textual Cultural Metadata – Tradition Code, Origin, Material
Artifact
Textual
FEATURE EXTRACTION
Image Feature Extraction CNN – Multi-Dimensional Feature Vectors
Cultural Metadata Embedding Layer – Text-Based Embedding Vectors
Image Feature
Embedding
Feature Fusion Module – Concatenation and Attention Fusion
Fused Feature Representation
Fused Feature Representation
Fused Feature Representation
CLASSIFICATION AND SCORING MODELS
Cultural Tradition
Authenticity Scoring
Buyer Recommendation Model (126)
MODEL
Tradition Code and Confidence Score
Composite Authenticity Score
Personalised Artifact

445 **Figure. 8**